

Problem of the Week 13

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Motivation

This week's Problem of the Week is focused on comparing and contrasting the supervised and unsupervised learning.

Completing this assignment

Fill in the cell as indicated below. When you are happy with your response, produce a clean rendition of your notebook. Then, create a PDF of your notebook and upload both documents (PDF and ipynb file) to Canvas.

Problem 13.1 - Differences Between Supervised and Unsupervised Learning

What is the main difference between supervised and unsupervised learning? Write 2-3 sentences to describe this difference. Use specific examples from lectures to illustrate your point.

Supervised learning already has a bunch of labeled data to train off of while unsupervised learning just kinda learns patterns from the data itself. We had labeled sea creature image data which is labeled versus something like the cloud coverage data that didn't come with labels to what is and isn't covered in clouds, they're just images.

Problem 13.2 - k-Nearest Neighbors vs k-Mean Clustering

Both of the algorithms k-Nearest Neighbors vs k-Mean Clustering use the Euclidean distance to compare similarities between data points. What is the key difference between these algorithms? Which one is unsupervised and which one is supervised? Explain in 2-4 sentences.

k-Nearest Neighbors is a supervised algorithm, while k-Means Clustering is unsupervised. the k in kNN is for the number of nearby points, while the k in k-Mean clustering refers to the number of clusters. kNN is meant to predict future labels for labeled data while k-Means clustering is supposed to just cluster existing data.

Problem 13.3 - Principal Component Analysis

In Principal Component Analysis, we use an algorithm to find patterns in our data without necessarily specifying what the patterns should be. In a similar sense, with neural networks, we also let our models learn patterns in the data. However, these are fundamentally different

approaches for investigating our data. Explain the key difference in terms of supervised vs unsupervised learning in 2-4 sentences.

PCA is an example of unsupervised learning since it takes disorganized data and sorts out the main things that create the variance in the data. This is compared to neural networks that require data to be labeled to run through its inputs to outputs.